

FinAccrual ERP Platform

Project Setup and Installation Steps

From a clean machine to a running FinAccrual environment

Item	Detail
Project	FinAccrual ERP Platform (Backend API, Excel Add-in, Admin Dashboard)
Document	Project Setup and Installation Steps

This guide explains how to install and run the Backend API, the Excel Add-in and the Admin Dashboard, how to verify the installation, and how to fix the most common problems.

1. Project Structure

The repository (ERP-main) is a monorepo with three applications and a few shared files.

```
ERP-main/  
├── Backend/Backend/    Node.js + Express 5 API (MySQL, Sequelize,  
                        JWT, OAuth)  
├── Frontend-React/     Microsoft Excel task-pane Add-in (React 19,  
                        Office.js, Webpack)  
├── Admin/              Admin Dashboard (React 19 + TypeScript, Vite)  
├── docker-compose.yml  Nginx container (optional load balancer)  
├── nginx.conf          Nginx routing to backend instances 8000 / 8001  
└── README.md
```

Application	Technology	Local URL
Backend API	Node.js, Express 5,	http://localhost:8000 (API under

	Sequelize, MySQL	/api)
Excel Add-in	React 19, Office.js, Webpack 5	https://localhost:3001/taskpane.html
Admin Dashboard	React 19, TypeScript, Vite, Redux Toolkit	http://localhost:5173
Nginx (optional)	nginx:alpine in Docker	http://localhost:8085

2. Prerequisites

Requirement	Version / Detail	Used for
Node.js	18 or newer (README minimum). Node 20 LTS or 22 LTS is recommended for the Admin app (Vite 8, ESLint 10, TypeScript 6)	All three applications
npm	9 or newer	Dependency installation
MySQL Server	8.x recommended (5.7+ works with Sequelize 6)	Backend database
Microsoft Excel	Desktop (Windows / macOS) or Excel on the web	Running the add-in
Git	Any recent version	Getting the source code
Docker Desktop	Optional	Nginx reverse proxy
Intuit developer account	developer.intuit.com	QuickBooks Online app and sandbox company
Xero developer account	developer.xero.com	Xero app and demo company
Google Cloud project	console.cloud.google.com	Google sign-in credentials
Microsoft Entra ID app	portal.azure.com	Microsoft sign-in credentials

Check the installed versions:

node -v

npm -v

mysql --version

3. Installation Steps

Step 1 — Get the source code

```
git clone <repository-url> ERP-main
cd ERP-main
```

If you received a ZIP file, extract it and open a terminal in the ERP-main folder.

Step 2 — Create the database

Log in to MySQL and create an empty database and user (tables are created automatically by the backend on first start).

```
CREATE DATABASE finaccrual CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;
CREATE USER 'finaccrual'@'localhost' IDENTIFIED BY '<strong-password>';
GRANT ALL PRIVILEGES ON finaccrual.* TO 'finaccrual'@'localhost';
FLUSH PRIVILEGES;
```

Step 3 — Register the OAuth applications

Create apps with Intuit, Xero, Google and Microsoft and note each client ID and secret. Register these redirect URIs for local work:

Provider	Redirect URI
QuickBooks	http://localhost:8000/api/quickbooks/callback
Xero	http://localhost:8000/api/xero/callback
Google	http://localhost:8000/api/auth/google/callback
Microsoft	http://localhost:8000/api/microsoft/callback

On macOS the backend runs on HTTPS in development (see Step 5), so the registered URIs must then start with <https://>.

Step 4 — Install and configure the Backend

```
cd Backend/Backend
npm install
cp .env.example .env      # Windows PowerShell: Copy-Item .env.example .env
```

Open `.env` and fill in the database settings, `JWT_SECRET`, and the QuickBooks, Xero, Google and Microsoft credentials. The full list of variables is described in the **Environment and Configuration Setup** document.

Step 5 — Start the Backend

```
npm run dev    # development, auto-restart with nodemon
# or
npm start      # plain node src/server.js
```

Successful start-up prints Database synchronized. followed by Node.js backend running on <http://localhost:8000>. Verify it:

```
curl http://localhost:8000/api/health
```

Expected response:

```
{ "success": true, "server": "healthy", "instance": "server-1", "database":  
"connected" }
```

macOS To let the HTTPS task pane call the backend, install the Office development certificates once (`npx office-addin-dev-certs install` inside Frontend-React). The backend then detects them in `~/.office-addin-dev-certs` and starts on <https://localhost:8000>.

Step 6 — Install and run the Excel Add-in

```
cd Frontend-React
npm install
npm run dev-server    # Webpack dev server on https://localhost:3001
```

Keep that terminal open. In a **second** terminal load the add-in into Excel:

```
cd Frontend-React
npm start          # office-addin-debugging: sideloads manifest.xml into desktop
Excel.
```

To sideload manually instead, use **Insert → Add-ins → My Add-ins → Upload My Add-in** in Excel and select `Frontend-React/manifest.xml`. Stop debugging with `npm run stop`.

If a certificate prompt appears, accept installing the local CA certificate. The dev server uses the certificates generated by office-addin-dev-certs.

Step 7 — Install and run the Admin Dashboard

Terminal A – two backend instances

```
cd Admin
```

```
npm install
```

```
npm run dev # Vite on http://localhost:5173 (proxies /api to :8000)
```

Step 8 — (Optional) Start Nginx with two backend instances

```
cd Backend/Backend
```

```
npm run dev:all # server-1 on 8000, server-2 on 8001
```

```
# Terminal B – Nginx on port 8085
```

```
cd ../../
```

```
Docker compose up -d
```

Requests to <http://localhost:8085/api/...> are forwarded to the two instances. See the sticky-session warning in the configuration document before using this for OAuth flows.

4. First-Run Verification Checklist

#	Check	Expected result
1	Open http://localhost:8000/api/health	server: healthy, database: connected
2	Check MySQL: SHOW TABLES;	users, admins, quickbooks_quickbookstoken, xero_tokens, notifications
3	Open Excel and start the FinAccrual add-in	Task pane loads with the sign-in screen
4	Sign in with Google or Microsoft	Pop-up closes; a row appears in users
5	Choose "Start Free Trial" (or a	users.plan is filled in

	plan)	
6	Connect a QuickBooks sandbox or Xero demo company	Row appears in the ERP token table with status Not Synced
7	Click Setup, then Pull Master Data	Sheets are created and filled; status becomes Active
8	Open http://localhost:5173	Admin dashboard login page appears

5. Running the Tests

The backend has Jest unit tests (mappers, Excel validation, OAuth helpers) and Supertest integration tests.

Command (in Backend/Backend)	Purpose
npm test	Run all tests once
npm run test:watch	Re-run tests on file change
npm run lint / npm run lint:fix	ESLint check / auto-fix
npm run format	Prettier formatting
npm run loadtest	Autocannon load test (scripts/loadtest.js)

Frontend tooling: npm run validate (in Frontend-React) checks the manifest; npm run lint (in Admin) runs ESLint.

6. Production Build and Deployment

6.1 Backend

```
cd Backend/Backend
```

```
npm ci --omit=dev
```

```
NODE_ENV=production PORT=8000 node src/server.js
```

- ☐ Provide all variables through the process environment or a protected .env.
- ☐ Run under a process manager (PM2, systemd, or a container) and put an HTTPS reverse proxy in front.
- ☐ Keep DB_SYNC_ALTER=false and back up the MySQL database regularly.

6.2 Excel Add-in

cd Frontend-React

API_BASE=https://api.finaccrual.com npm run build

- ☐ Upload the generated dist/ folder to a static HTTPS host (for example <https://addin.finaccrual.com>).
- ☐ Distribute manifest.prod.xml through the Microsoft 365 admin centre (Integrated Apps) or AppSource.

6.3 Admin Dashboard

cd Admin

npm run build # type-check + Vite build into dist/

- ☐ Host dist/ on any static server. Replace the hard-coded <http://localhost:8000> URLs in the source before building for another host.

7. Troubleshooting

Symptom	Likely cause	Fix
Unable to connect to the database at start-up	MySQL not running, wrong DB_* values, or database missing	Start MySQL, verify credentials, run CREATE DATABASE
Access denied for user	Wrong DB_USER / DB_PASSWORD	Correct the values in .env; confirm the user has privileges on the schema
/api/health shows database: disconnected	Database dropped after start-up	Check MySQL, restart the backend
redirect_uri_mismatch (Google / Intuit / Xero)	Redirect URI in the provider portal differs from .env	Make both identical, including protocol and port
Sign-in pop-up blocked	Browser or Excel blocks pop-ups	Allow pop-ups for the add-in
Session expired. Please reconnect QuickBooks/Xero	Session cookie lost between connect and callback (HTTP/HTTPS mismatch, or callback hit another instance)	Use consistent protocol; use a single instance or sticky sessions
Task pane blank or	Dev certificates missing	Run npx office-addin-

certificate warning	or untrusted	dev-certs install and restart Excel
macOS add-in cannot reach API	Mixed content: HTTPS page calling HTTP API	Install dev certs so the backend runs on HTTPS
Please check your network connection banner	Backend not running or wrong API_BASE	Start the backend; rebuild the add-in with the correct API_BASE
ERR_SESSION_EXPIRED after about an hour	Access token lifetime (1 hour) reached	Sign in again (see the authentication document)
npm install fails on bcrypt or sqlite3	Native module build problem	Use a supported Node LTS version and install build tools (Windows Build Tools, Xcode CLT, or build-essential)
Port 8000 / 3001 / 5173 already in use	Another process on the port	Stop it or change PORT (backend) / Webpack port / Vite --port

8. Command Quick Reference

Folder	Command	What it does
Backend/Backend	npm install	Install dependencies
Backend/Backend	npm run dev	Start API with nodemon
Backend/Backend	npm run dev:all	Start two instances (8000, 8001)
Backend/Backend	npm start	Start API (production style)
Backend/Backend	npm test	Run Jest tests
Frontend-React	npm run dev-server	Webpack dev server on https://localhost:3001
Frontend-React	npm start / npm run stop	Sideload / unload the add-in in Excel
Frontend-React	npm run build	Production bundle in dist/
Admin	npm run dev	Vite dev server on port 5173
Admin	npm run build	Type-check and build
ERP-main	docker compose up -d	Start Nginx on port 8085

